Balamurugan S

Cyber Security
balamar2007@gmail.com | +91 8508576159 | Coimbatore | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Cyber Security student with a strong interest in ethical hacking, penetration testing, and Capture the Flag (CTF) challenges. Enthusiastic about hackathons, solving real-world cybersecurity problems, and building practical security solutions. Experienced in full-stack development and committed to improving technical skills through hands-on projects and competitive challenges.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C
- **Networking:** TCP/IP, DNS, Routing & Switching, CIDR & Subnetting, VPC Networking, Load Balancing, VPN, Firewalls, IDS/IPS, ACLs, Packet Analysis, Network Troubleshooting
- **Cybersecurity:** Penetration Testing, Web Application Security, Vulnerability Assessment, Binary Exploitation, Reverse Engineering, Digital Forensics, Malware Analysis, Cryptography, Container Security, Security Operations Center (SOC)
- **Security Tools:** Burp Suite, Metasploit, Nmap, Wireshark, Ghidra, GDB, pwndbg, Wazuh, OWASP ZAP
- **Platforms & Version Control:** Linux, Windows, Git, GitHub

PROJECTS

HostelFix – Smart Hostel Monitoring System

- Developed a full-stack hostel management platform with role-based dashboards for admin, warden, student, security and technicians.
- Implemented JWT authentication, role-based access control, PostgreSQL integration, and real-time dashboard updates using Socket.IO.
- Built digital gate-pass management with QR verification, digital signature, and laundry management.
- Integrated live announcements via Telegram bot notifications, along with AI-based CCTV monitoring featuring face recognition and fire/smoke detection.
- **Tech:** Node.js, Express.js, PostgreSQL, JavaScript, Python, YOLO, OpenCV, JWT, Socket.IO, Telegram Bot API

Smart Exam Hall Monitoring System

- Developing a smart exam monitoring system that integrates Veyon for remote student screen monitoring and centralized classroom supervision.
- Built a custom monitoring agent that captures client-side activity logs, including application switching, process execution, and clipboard activity.
- Integrated the agent with a real-time master dashboard so administrators can view student activity and collected logs.
- Implemented Supabase to store monitoring logs for centralized access and review.
- **Tech:** Veyon, Node.js, React.js, JavaScript, Socket.IO, Supabase, Python

ACHIEVEMENTS

- **Cyber.I@ne'26 2.0 – 1st Prize:** 8-hour CTF (AURA) at a national-level technical symposium organized by Mahendra Engineering College, Namakkal.
- **HACKNEXT'26 – Series 1.0 – 2nd Prize:** AI-powered 8-hour hackathon organized by SNS College of Technology, Coimbatore.
- **SUSTAINVERSE 2K26 – 2nd Prize:** 24-hour hackathon organized by DMI College of Engineering, Chennai.
- **Lun4R CTF 2026 – 29th Place:** National-level CTF organized by ROOT RIET at Rajadhani Institute of Engineering and Technology (RIET), Thiruvananthapuram, Kerala.
- **NULL ORIGIN CTF 2026 – 57th Place:** CTF organized by Team CyberXoX × CyberHX.

EDUCATION

B.E. in CSE (Cyber Security) (2024 – 2028)
Sri Shakthi Institute of Engineering and Technology, Coimbatore, TN
- CGPA: 6.52

Higher Secondary Certificate (HSC) (2023 – 2024)
Sri Kalaimagal Vidhyalaya Matric and Higher Secondary School, TN
- Score: 75.2%

Secondary School Leaving Certificate (SSLC) (2021 – 2022)
Sri Kalaimagal Vidhyalaya Matric and Higher Secondary School, TN
- Score: 65.6%

CERTIFICATIONS

- Introduction to Software Testing – Simplilearn
- Software Testing with Generative AI – Simplilearn